

STRATHALLEN, United Kingdom

WMO: 031440

Lat: 56.3264N

Lon: 3.7286W

Elev: 35

StdP: 100.91

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.9	(c) -4.5	(d) -7.5	(e) 2.0	(f) -3.7	(g) -5.7	(h) 2.3	(i) -1.7	(j) 13.4	(k) 7.4	(l) 12.0	(m) 8.3	(n) 0.6	(o) 240	(p) 0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	22.9	17.4	21.1	16.2	19.7	15.3	18.2	21.6	17.1	19.9	16.2	18.7	3.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.1	19.7	15.9	11.3	18.4	15.0	10.7	17.6	51.6	21.7	48.2	20.0	45.5	18.7	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.1	8.0	-9.3	25.8	2.9	2.0	-11.4	27.2	-13.1	28.4	-14.7	29.5	-16.8	30.9		
			-9.1	20.5	3.4	1.8	-11.5	21.8	-13.5	22.8	-15.4	23.8	-17.8	25.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	3.4	3.9	5.4	7.7	10.2	12.9	14.9	14.5	12.5	9.2	5.9	3.4
	DBStd	5.05	3.36	3.13	2.91	2.64	2.73	2.20	2.14	2.15	2.30	2.89	3.32	4.17
	HDD10.0	1021	204	171	146	78	31	3	0	1	6	49	128	206
	HDD18.3	3526	462	404	402	320	253	163	109	119	176	282	374	463
	CDD10.0	539	0	1	2	8	36	91	151	141	81	25	4	1
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	37	0	0	0	0	2	6	20	9	0	0	0	0
	CDH26.7	2	0	0	0	0	0	1	1	0	0	0	0	0
	WSAvg	3.2	3.6	3.7	3.8	3.4	3.1	3.0	2.7	2.7	2.9	3.1	3.1	3.2
(15) Precipitation	PrecAvg	996	124	80	62	61	70	67	78	83	66	102	99	106
	PrecMax	1282	281	190	137	117	133	120	154	188	152	177	195	281
	PrecMin	640	32	9	12	17	22	23	17	22	13	10	32	27
	PrecStd	145	61	45	29	26	28	29	37	42	36	48	42	61
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	12.6	14.7	18.3	22.6	23.6	25.7	24.7	21.4	17.2	14.3	12.9
		MCWB	10.4	10.4	10.2	12.3	16.4	17.4	19.0	19.6	16.3	14.3	12.3	11.3
	2%	DB	10.7	11.2	12.5	15.3	19.3	20.9	23.1	22.0	19.2	15.7	13.0	11.6
		MCWB	9.2	9.3	9.3	10.7	14.0	15.8	17.8	17.0	15.1	13.4	11.2	10.2
	5%	DB	9.6	10.1	11.2	13.8	17.1	19.2	21.3	20.2	17.9	14.7	11.9	10.5
		MCWB	8.1	8.4	8.6	10.1	12.4	14.7	16.7	15.9	14.2	12.6	10.3	9.2
	10%	DB	8.5	8.9	10.0	12.5	15.3	17.7	19.5	18.9	16.7	13.6	10.7	9.2
		MCWB	7.0	7.1	7.8	9.2	11.4	13.8	15.6	15.0	13.6	11.7	9.2	7.9
	0.4%	WB	10.7	11.0	11.2	13.2	17.1	18.5	20.4	20.0	17.6	14.9	12.9	11.7
		MCDB	11.8	12.1	13.8	16.5	21.8	22.3	23.8	23.9	20.2	16.2	13.9	12.7
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	9.5	9.7	10.0	11.7	14.7	16.5	18.5	17.8	16.1	13.9	11.6	10.5
		MCDB	10.6	10.9	11.8	14.5	18.5	19.7	22.2	20.7	18.2	15.3	12.6	11.5
	5%	WB	8.3	8.6	9.0	10.6	13.2	15.4	17.3	16.7	15.0	13.0	10.5	9.3
		MCDB	9.5	10.0	10.7	13.2	16.3	18.3	20.3	19.1	17.0	14.3	11.6	10.4
	10%	WB	7.2	7.4	8.1	9.6	12.0	14.3	16.3	15.9	14.2	12.1	9.5	8.1
		MCDB	8.4	8.8	9.8	12.0	14.6	16.9	18.8	18.0	16.0	13.4	10.6	9.2
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	6.1	6.9	7.9	8.9	8.5	8.7	8.8	8.0	6.9	6.1	5.7
		MCDBR	5.9	6.6	8.4	10.3	12.2	11.5	11.8	10.8	9.8	7.4	6.5	6.4
		MCWBR	5.3	5.3	5.8	6.5	7.3	7.2	6.9	6.4	6.2	5.5	5.6	5.8
	5% WB	MCDBR	5.9	5.9	7.3	9.2	10.7	10.5	10.2	9.3	8.0	6.7	6.2	6.3
		MCWBR	5.6	5.2	5.6	6.2	6.9	7.1	6.7	6.2	5.7	5.4	5.7	5.9
(40) Clear-Sky Solar Irradiance	taub		0.321	0.336	0.366	0.388	0.390	0.388	0.388	0.371	0.358	0.341	0.315	0.304
	taud		2.222	2.319	2.270	2.247	2.290	2.325	2.330	2.385	2.430	2.411	2.310	2.195
	Ebn at Noon		554	718	786	826	849	855	845	837	795	707	568	470
	Edh at Noon		56	75	100	118	120	118	115	102	85	68	51	45
(44) All-Sky Solar Radiation	RadAvg		0.48	1.13	2.15	3.54	4.71	4.62	4.44	3.54	2.48	1.33	0.63	0.35
	RadStd		0.07	0.12	0.25	0.35	0.33	0.42	0.41	0.26	0.17	0.16	0.06	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page